

HRW 12e Conceptual Practice Quiz — Student Version

Multiple choice, one correct answer per question.

EM Induction (Q1–Q10)

Q1. What does Lenz's law state about the direction of an induced current?

- (a) It enhances the change in flux
- (b) It is independent of flux changes
- (c) It always flows clockwise
- (d) It opposes the change in flux that produced it

Q2. According to Faraday's law, the magnitude of the induced emf in a loop is proportional to:

- (a) The magnetic field strength only
- (b) The loop resistance
- (c) The rate of change of magnetic flux
- (d) The loop circumference

Q3. Which expression correctly defines magnetic flux through a flat loop area A in field B at angle θ ?

- (a) $\Phi_B = BA$
- (b) $\Phi_B = B/A$
- (c) $\Phi_B = BA \cos \theta$
- (d) $\Phi_B = BA \sin \theta$

Q4. In motional emf, charges in a conductor moving with velocity $\vec{v}$ through field $\vec{B}$ are separated by which force?

- (a) $q\vec{E}$
- (b) Gravity
- (c) $q\vec{v} \times \vec{B}$
- (d) Nuclear force

Q5. A loop enters a region of increasing magnetic field into the page. To oppose the change, the induced current's own magnetic field points:

- (a) Into the page
- (b) Out of the page
- (c) Horizontally
- (d) It is zero

Image: images/em_induction_loop.png

Q6. The negative sign in Faraday's law $\mathcal{E} = -d\Phi_B/dt$ expresses:

- (a) A meaningless sign convention
- (b) That emf is always negative
- (c) Lenz's law: induced emf opposes flux changes
- (d) That flux is always decreasing

Q7. Eddy currents are:

- (a) DC currents in batteries
- (b) Loops of induced current in bulk conductors
- (c) Only currents in superconductors
- (d) Hypothetical with no real effects

Q8. Which situation will NOT induce an emf in a conducting loop?

- (a) Changing the magnetic field through the loop
- (b) Changing the loop's area in a magnetic field
- (c) Rotating the loop in a magnetic field
- (d) Keeping field, area, and orientation all constant

Q9. A transformer operates primarily due to:

- (a) Coulomb's law
- (b) Faraday's law of induction
- (c) Gauss's law for magnetism
- (d) Ohm's law

Q10. The SI unit of magnetic flux is:

- (a) Tesla
- (b) Weber
- (c) Henry
- (d) Volt

LR Circuits & Inductors (Q11–Q20)

Q11. In an RL circuit, the time constant τ is:

- (a) R/L
- (b) L/R
- (c) RC
- (d) $1/RC$

Q12. Immediately after an RL circuit is connected to a DC source, the ideal inductor:

- (a) Allows full current instantly
- (b) Opposes any change in current, acting like an open circuit

- (c) Behaves like a capacitor
- (d) Has no effect

Q13. Long after connection to a DC source, the ideal inductor in an RL circuit behaves like:

- (a) An open circuit
- (b) A short circuit (wire)
- (c) A large resistor
- (d) A capacitor

Q14. The energy stored in an inductor is:

- (a) LI
- (b) $\frac{1}{2}LI^2$
- (c) $\frac{1}{2}CV^2$
- (d) RI^2

Q15. The SI unit of inductance is:

- (a) Tesla
- (b) Weber
- (c) Henry
- (d) Ohm

Q16. The induced emf across an inductor is:

- (a) $\mathcal{E} = LI$
- (b) $\mathcal{E} = -L dI/dt$
- (c) $\mathcal{E} = IR$
- (d) $\mathcal{E} = qvB$

Q17. Increasing L (with R fixed) in an RL circuit makes the current reach steady state:

- (a) Faster
- (b) More slowly
- (c) At the same rate
- (d) Oscillatory

Q18. Placing a ferromagnetic core inside a coil generally:

- (a) Decreases its inductance
- (b) Increases its inductance
- (c) Leaves inductance unchanged
- (d) Eliminates inductance

Q19. In the RL circuit diagram, which element stores energy in a magnetic field?

- (a) Battery
- (b) Resistor
- (c) Inductor
- (d) Switch

Image: images/rl_circuit_diagram.png

Q20. Self-inductance is:

- (a) Induction of emf in a coil by its own changing current
- (b) Induction only between two separate coils
- (c) A property unique to resistors
- (d) A purely electrostatic effect

AC Circuits I (Q21–Q30)

Q21. A standard AC source voltage is usually written as:

- (a) $V(t) = V_0 t$
- (b) $V(t) = V_0 \sin(\omega t)$
- (c) $V(t) = IR$
- (d) $V(t) = V_0 e^t$

Image: `images/ac_waveform.png`

Q22. The rms value of $V(t) = V_0 \sin(\omega t)$ is:

- (a) V_0
- (b) $V_0/2$
- (c) $V_0/\sqrt{2}$
- (d) $\sqrt{2}V_0$

Q23. In a purely resistive AC circuit, current and voltage are:

- (a) In phase
- (b) 90° out of phase
- (c) 180° out of phase
- (d) Unrelated

Q24. The inductive reactance is:

- (a) $X_L = 1/(\omega C)$
- (b) $X_L = \omega L$
- (c) $X_L = R$
- (d) $X_L = 1/R$

Q25. The capacitive reactance is:

- (a) $X_C = 1/(\omega C)$
- (b) $X_C = \omega C$

- (c) $X_C = R$
- (d) $X_C = \omega L$

Q26. In a purely inductive AC circuit, current:

- (a) Leads voltage by 90°
- (b) Lags voltage by 90°
- (c) Is in phase
- (d) Is zero

Q27. In a purely capacitive AC circuit, current:

- (a) Leads voltage by 90°
- (b) Lags voltage by 90°
- (c) Is in phase
- (d) Is opposite in phase

Q28. Impedance Z in AC circuits is:

- (a) Only the resistance R
- (b) The complex combination of resistance and reactance
- (c) Always purely imaginary
- (d) Always purely real

Q29. A phasor diagram is primarily used to represent:

- (a) Static electric fields
- (b) Phase relationships between AC quantities
- (c) Circuit schematics
- (d) Particle trajectories

Q30. Rms values are useful because they:

- (a) Equal peak values
- (b) Directly relate to average power like DC values
- (c) Remove phase information
- (d) Are easier to graph

AC Circuits II (Q31–Q40)

Q31. In a series RLC circuit at resonance, the net reactance is:

- (a) Positive
- (b) Negative
- (c) Zero
- (d) Infinite

Image: images/rlc_resonance.png

Q32. At resonance in a series RLC circuit, the current for a given source is:

- (a) Minimum
- (b) Maximum
- (c) Zero
- (d) Undefined

Image: images/rlc_resonance.png

Q33. The resonant angular frequency ω_0 of a series RLC circuit is:

- (a) $1/RC$
- (b) $1/\sqrt{LC}$
- (c) R/L
- (d) $\sqrt{LC}$

Image: images/rlc_resonance.png

Q34. The quality factor Q of a resonance describes:

- (a) Only resistance
- (b) Sharpness (selectivity) of the resonance peak
- (c) Circuit size
- (d) Battery voltage

Image: images/r1c_resonance.png

Q35. A power factor near 1 indicates:

- (a) Mostly reactive power
- (b) Mostly real (useful) power
- (c) No power delivered
- (d) Infinite impedance

Q36. Capacitors are added to inductive loads mainly to:

- (a) Increase frequency
- (b) Correct (improve) power factor
- (c) Increase resistance
- (d) Eliminate real power

Q37. In an ideal transformer, V_s/V_p equals:

- (a) N_p/N_s
- (b) N_s/N_p
- (c) I_p/I_s
- (d) I_s/I_p

Q38. For an ideal transformer, neglecting losses:

- (a) Input power equals output power
- (b) Input power is twice output power
- (c) Output power is zero
- (d) Power is not conserved

Q39. Above resonance, a series RLC circuit behaves mainly:

- (a) Capacitive
- (b) Inductive
- (c) Resistive
- (d) Superconducting

Q40. Below resonance, a series RLC circuit behaves mainly:

- (a) Capacitive
- (b) Inductive
- (c) Resistive
- (d) Superconducting

Maxwell's Equations & EM Waves (Q41–Q50)

Q41. Gauss's law for magnetism states:

- (a) $\nabla \cdot \vec{B} = \mu_0 \rho_m$
- (b) Net magnetic flux through any closed surface is zero
- (c) Magnetic field lines start on magnetic charges
- (d) Magnetic field is always uniform

Q42. In vacuum, the speed of light is:

- (a) $c = \mu_0 \epsilon_0$
- (b) $c = 1/(\mu_0 \epsilon_0)$

(c) $c = 1/\sqrt{\mu_0\epsilon_0}$

(d) $c = \sqrt{\mu_0/\epsilon_0}$

Q43. In an electromagnetic plane wave:

(a) $\vec{E} \parallel \vec{B} \parallel$ propagation

(b) $\vec{E} \perp \vec{B}$ and both $\perp$ propagation

(c) $\vec{E}$ only along propagation

(d) $\vec{B}$ only along propagation

Image: images/em_wave.png

Q44. The displacement current term was added to Ampere's law to:

(a) Change Gauss's law

(b) Match Faraday's law

(c) Ensure consistency with charge conservation in time-varying fields

(d) Remove magnetic fields

Q45. The Poynting vector $\vec{S}$ represents:

(a) Charge density

(b) Energy flux (power per unit area) of EM waves

(c) Magnetic monopole current

(d) Phase difference

Q46. Electromagnetic waves carry:

(a) Only electric energy

(b) Only magnetic energy

(c) Both energy and momentum

(d) Neither

Q47. For a plane EM wave in free space, E_0/B_0 equals:

- (a) 1
- (b) c
- (c) $1/c$
- (d) 0

Q48. A linearly polarized EM wave is one in which:

- (a) Frequency changes with time
- (b) The electric field oscillates along a fixed direction
- (c) Direction of propagation changes randomly
- (d) The magnetic field is zero

Q49. Most radio-frequency EM waves are generated by:

- (a) Static charges
- (b) Accelerating charges or oscillating currents
- (c) Permanent magnets
- (d) Nuclear reactions only

Q50. The diagram showing perpendicular $\vec{E}$ and $\vec{B}$ traveling together represents:

- (a) Sound wave
- (b) Water wave
- (c) Electromagnetic plane wave
- (d) Standing wave on a string

Image: [images/em_wave.png](#)

Reflection & Refraction (Q51–Q60)

Q51. Snell's law is:

- (a) $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- (b) $F = ma$
- (c) $V = IR$
- (d) $pV = nRT$

Q52. When light enters a medium with higher n , it:

- (a) Speeds up and bends away from the normal
- (b) Slows down and bends toward the normal
- (c) Slows down and bends away from the normal
- (d) Is unchanged

Q53. Total internal reflection occurs when light:

- (a) Goes from low n to high n
- (b) Goes from high n to low n with $\theta > \theta_c$
- (c) Strikes any mirror
- (d) Travels in vacuum

Q54. For a plane mirror, the angle of reflection equals:

- (a) Angle of refraction
- (b) Angle of incidence
- (c) Critical angle
- (d) 90°

Q55. The index of refraction is:

- (a) $n = v/c$
- (b) $n = c/v$
- (c) $n = 1/v$
- (d) $n = c$

Q56. If $n_2 < n_1$, the refracted ray:

- (a) Bends toward the normal
- (b) Bends away from the normal
- (c) Always totally internally reflects
- (d) Disappears

Q57. At Brewster's angle, the reflected light is:

- (a) Unpolarized
- (b) Circularly polarized
- (c) Linearly polarized
- (d) Completely absorbed

Q58. A ray entering along the normal ($\theta = 0$) into a new medium:

- (a) Bends toward the surface
- (b) Bends away from the surface
- (c) Continues straight
- (d) Totally reflects

Q59. A correct refraction diagram from air to glass shows the refracted ray:

- (a) Bending toward the normal
- (b) Bending away from the normal
- (c) Parallel to the surface
- (d) Returning to air

Image: images/refraction_diagram.png

Q60. Angles of incidence and refraction are always measured:

- (a) From the surface
- (b) From the normal
- (c) From the vertical
- (d) Arbitrarily